

Project Proposal

GreenBite: A Sustainable and Healthy Meal Recommendation Platform for Corporate Cafeterias

1. Project Overview

HachimiSoft, a provider of enterprise digital solutions, proposes **GreenBite**—a sustainable and healthy meal recommendation platform—to help large enterprises optimize employee diet management and cafeteria operations, targeting the ByteDance corporate cafeteria as a potential client.

As tech companies expand, flexible working hours and diverse dietary needs (e.g., fat loss, high protein, low calorie) make daily cafeteria demand inherently unpredictable. Traditional cafeterias often fail to address these personalized needs, driving employees toward less healthy, packaging-heavy external food deliveries. GreenBite aims to solve these issues, reduce single-use waste, and boost cafeteria utilization through intelligent meal recommendations, inventory management, and data analysis.

2. Problem Background and Client

ByteDance provides extensive corporate cafeteria services but faces two primary challenges. First, unpredictable employee dining behaviors—caused by meetings, overtime, or remote work—make demand forecasting difficult. Traditional experience-based procurement often leads to severe food waste when employees dine elsewhere, or conversely, causes popular meals to sell out early.

Second, employees lack the nutritional transparency needed to make healthy choices. This data gap drives reliance on external food delivery apps, which lowers cafeteria utilization and negatively impacts employee health. Furthermore, without robust data to track meal popularity and ingredient waste, managers struggle to optimize procurement, thereby increasing operational costs. GreenBite serves as the ideal digital upgrade solution for these operational bottlenecks.

3. Solution

GreenBite provides ByteDance with an intelligent meal recommendation and inventory management system. Employees can view daily menus and receive personalized recommendations based on health goals and transparent tags (e.g., low-calorie). This makes healthy choices easier, boosts cafeteria attractiveness, and reduces reliance on food delivery.

To actively reduce food waste, the system combines inventory and historical dining data to prioritize meals using overstocked or near-expiry ingredients. Additionally, a comprehensive data analysis dashboard provides managers with real-time insights into meal popularity, inventory usage, and waste risks, empowering them to optimize procurement strategies.

4. Technical Implementation

GreenBite will adopt React, Spring Boot, MyBatis, PostgreSQL, and Docker as its core technology stack.

- **Frontend (React):** Chosen for its component-based approach, React perfectly suits highly interactive UIs like dynamic menus, preference pages, and manager dashboards, offering high development efficiency and maintainability.

- **Backend (Spring Boot):** Selected for building scalable RESTful Web services. It provides a mature framework to stably handle recommendation logic, inventory management, and easily implement role-based access control for different users (employees, staff, administrators).
- **Data Access (MyBatis):** MyBatis handles database interactions by granting developers precise control over complex SQL statements. This is crucial for business-critical operations like inventory updates and statistical analysis, improving both query efficiency and maintainability.
- **Database (PostgreSQL):** A highly stable relational database used to store user profiles, meal data, and inventory. Its robust support for multi-table joins and complex queries makes it ideal for recommendation analysis and report generation.
- **Architecture & Deployment:** GreenBite adopts a frontend-backend separation model to improve modularity and enable parallel development. Docker is utilized to containerize the application, eliminating environment discrepancies and facilitating seamless deployment and testing in virtual machine environments.

5. Sustainability Contribution and SDG Alignment

GreenBite enhances sustainability by reducing food waste and promoting employee health. The system reduces ingredient waste through inventory-aware recommendations and optimizes procurement strategies via data analysis. Concurrently, it encourages healthier meals through personalization, reducing reliance on delivery and lowering single-use packaging waste.

This aligns perfectly with UN Sustainable Development Goals **SDG 3 (Good Health and Well-being)** by improving employee dietary structures, and **SDG 12 (Responsible Consumption and Production)** by cutting food waste through inventory optimization, thereby boosting the cafeteria's sustainable operational capacity.

6. Ethical Issues and Social Impact

GreenBite heavily prioritizes data privacy and algorithmic ethics. The system strictly adheres to data minimization, collecting only necessary preferences, and uses encrypted storage and role-based access control to protect user privacy. To avoid algorithmic bias, the recommendation system remains transparent, allowing administrators to adjust weights to genuinely encourage healthy dietary choices.

Socially, GreenBite helps employees form healthy dietary habits, improves cafeteria operational efficiency, and significantly reduces single-use packaging and food waste, ultimately elevating the enterprise's social responsibility image.

7. Expected Outcome

Upon completion, HachimiSoft will deliver a complete, browser-accessible GreenBite platform to ByteDance, supporting personalized recommendations, inventory management, and data analysis. The system is expected to drastically reduce ingredient waste, increase cafeteria utilization, and improve employee health. Through a data-driven approach, GreenBite will establish a highly efficient, sustainable operational model with strong potential for expansion to other corporate or campus cafeterias.